# FINAL PROJECT REPORT

**Submitted by:**

**VINH TRAN**

**CWID: 893549451**

**California State University, Fullerton**

**CPSC-597**

**Fall 2026**

## Advisor: Dr. Paul Salvador Inventado

**Department of Computer Science**

**California State University Fullerton**

CALIFORNIA STATE UNIVERSITY
FULLERTON™

# Department of Computer Science

This project has been satisfactorily demonstrated and is of suitable form

This project report is acceptable in partial fulfilment of the requirements for the award of Master of Science degree in Computer Science.

**A Web Application Integrating an AI-Based Intrusion Detection System for Network Traffic Analysis**

Project Title

**Vinh Tran**

Student Name

**Dr. Paul Salvador Inventado**

Advisor's Name

Advisor's signature Date

Reviewer's name *(if you have one)*

Reviewer's signature Date

# Table of Contents

**Abstract** ..... 5
**1. Introduction** ..... 6
**1.1 Motivations** ..... 7
**1.2 Project Objectives** ..... 7
**1.3 Contributions** ..... 8
**2. Literature Review** ..... 9
**2.1 Artificial Intelligence (AI)** ..... 9
**2.1.1 Background Information and Development of AI** ..... 9
**2.1.2 The Role of AI in Network Security Technologies** ..... 9
**2.2 External Threats** ..... 11
**2.2.1 Background Information** ..... 11
**2.2.2 The Application Models of AI in Network Security Technologies Solve External Threats** 11
**2.3 Internal Threats** ..... 12
**2.3.1 Background Information** ..... 12
**2.3.2 The Application Models of AI in Network Security Solve Internal Threats** ..... 14
**2.4 Challenges of AI in Network Security** ..... 15
**3. Methodology** ..... 17
**3.1 Dataset** ..... 17
**3.1.1 Overview** ..... 17
**3.1.2 Preprocessing and Normalizing Dataset** ..... 17
**3.2 Machine Learning Models** ..... 19
**3.3 Server Development** ..... 20
**3.4 Client Development** ..... 20
**4. Results and Discussion** ..... 21
**5. Conclusion** ..... 22
**5.1 Future Work** ..... 22
**5.2 Limitations** ..... 22
**6. Reference** ..... 23

## Table of Figures

Figure 1 .....5
Figure 2 .....18

## List of Tables

Table 1 .....17
Table 2 .....19
Table 3 .....19

# Abstract

With the development of technologies in the information of age, digital transformation is considered a popular method that most individuals and enterprises choose to format the data and store them into their computer or cloud services. Even though this modern technology brought new strategies in converting the status and accessibility of data from physical to publicly available on the internet, it also caused the risks from cybersecurity. That means, sensitive data becomes the vulnerable target to internal and external threats through communicating network systems. There existed some traditional security tools like Intrusion Detection System (IDS) that could detect, identify, and protect the data; however, it became useless the new and sophisticated attacks. These types of threats could easily bypass the detection and perform malicious actions. To solve its limitations, the purpose of this project is to present the possible solution by developing an AI-based Intrusion Detection System, which operates as a web application. The prototype system will take network traffic information as inputs, analyze their data, and classify them into different categories based on their activities and behaviors. Moreover, the system will train multiple supervised learning models on the latest and reality UNSW-NB15 dataset. After training the algorithms, Extreme Gradient Boost (XGBoost) is the classification model that achieved the accuracy score at 76.6% and was selected to become the core one to analyze the incoming network traffic data. This project builds the server by using Flask in Python to bridge between the users and the system. To interact with the enhanced system, the users can utilize the user interface to input their own network traffic data, receive real-time classifications through the analysis visualized on the dashboard. Indeed, the system aims to manifest the enhancement of the AI-based IDS in detection accuracy and ability of monitoring different threats.

# 1. Introduction

In the Information Age, the rapid development of new techniques and methods drives corporations to innovate through the digital transformation of their data and information. Yet, this concurrently exposes their data to critical concerns related to network security. Mir, Wani, and Ibrahim (2013) introduced in their study that data is a high priority and importance for most organizations. It needs to be managed and secured because it contains important and sensitive information such as transaction records, intellectual property, or personal information related to customers and employers. Traditionally, to perform data management and protection, organizations secured information by storing it in physical storage. Later, data was digitalized by converting physical documents into digital files saved on computers or uploaded to cloud services. At first, this transformation seemed effective since it provided the convenience of data access while also limiting unauthorized access to sensitive information. However, by applying this method, corporations and organizations encountered five major classifications of concerns towards their information: “Theft”, “Fraud”, “Malicious Damage”, “Incompetence & mistakes”, and “Accidents & disasters” (Mir, Wani, and Ibrahim, 2013). As a result, data and information in large corporations and organizations become vulnerable targets of these internal and external threats which result in data breaches and impact on files, service continuity, and finance. Nevertheless, these concerns may be prevented and solved thanks to the development of Artificial Intelligence. Applying artificial intelligence in network security can enhance existing technologies making them more powerful and intelligent. Consequently, these technologies could perform their operations with higher efficiency and accuracy in data protection, threat detection, and cost reduction (Li, Zhang, Wang, & Xiao, 2023).

## 1.1 Motivations

Intrusion Detection System (IDS), for example, is a traditional security tool used for monitoring network traffic and detecting malicious activities. It will raise warnings or alerts to controllers if it identifies any potential anomalies. However, the tool may only work well towards specified threats in the database or system. Due to its weakness, attackers create new and more sophisticated ones so that their attacks cannot be detected by the system. Therefore, encountering new and sophisticated attacks from modern technologies makes the system useless, weaker, and less efficient. The simple solution is to constantly update the detection rules for the system, but for a long time, it is time-consuming and requires a lot of effort to do so. As a result, this project is to overcome the limitations of the system and provide another effective solution. By applying machine learning models of artificial intelligence, the intrusion detection system will become automated and robust. That means, the system takes and analyzes the network traffic patterns as training data. Then, when there are incoming traffic and activities, it can automatically monitor, identify, and update new types of attacks into its database.

## 1.2 Project Objectives

The main objective of the project is to manifest the efficiency, automation, and practical feasibility of an AI-powered Intrusion Detection System (IDS) for analyzing network traffic. In order to acquire the goal, the project will construct the prototype system that leverages machine learning, such as supervised learning models and deep learning to train and analyze the available datasets. By doing so, they enable the system to classify network traffic and detect the intrusions and anomalies.

The prototype is implemented as the web application that provides an interactive service for users to the system. Besides using and processing the available datasets, the project also accepts the datasets from users. That means, users allow to input their own traffic data, such as

uploading their appropriate files or manually entering network traffic flow information. After that, the system will return the results by presenting all necessary information (e.g., alerts, warnings, classifications, and model outputs) to users through the dashboard.

Overall, through the prototype system in this project, we can evaluate the performance of multiple machine learning models and compare their capabilities on detecting threats. We may also see that the detection in the AI-based IDS is automated and enhanced on accuracy. Finally, the project visualizes how the AI-based system is able to detect both known and unknown threats from real-world network environments.

## 1.3 Contributions

The project is to mainly connect the theory of applying machine learning models into the system by practically delivering the fully functional, AI-based intrusion detection system through the web application. Therefore, the contributions will include the following components:

- **Processed Dataset:** Selecting an appropriate dataset (i.e., UNSW-NB15) to analyze and preprocess so that it becomes the training and testing data for machine learning models.
- **Machine Learning Models Development:** Training multiple supervised learning models by using the processed dataset as the input data to analyze, evaluate, and compare their performance and accuracy. Using the metrics like precision, f1-score, and recall to determine the appropriate model to develop and apply into the intrusion detection system.
- **AI-based Intrusion Detection System:** Demonstrating that the system can automatically monitor, analyze, and identify both known attacks (from training data) and unknown threats (from the network traffic data collected by user's input).
- **Server Development:** Bridging the gap between the user and the intrusion detection system by handling requests and responses from both users and machine learning models.

- **Client Development:** Creating an interactive environment for users where they can input their network traffic information and receive the analysis as the classifications from the developed AI-based intrusion detection system.

# 2. Literature Review

## 2.1 Artificial Intelligence (AI)

### 2.1.1 Background Information and Development of AI

Artificial Intelligence is becoming more noticeable under the world's growth in information technology. Huang (2024) explained the term of artificial intelligence is denoted as the simulation of the human brain. Its main functions are to analyze, process, and perform tasks on computers like human experts. The timeline of research and development can be traced back to the period of 1950s. Initially, it was developed to logically solve problems, yet it struggled with complex ones due to computational limitations. In the period of 1980s, it marked the rise of expert systems in AI. They have the ability of performing decision-making tasks like humans to solve complex problems. For example, in medical, expert systems could diagnose a disease with high accuracy. After several decades, from the 1990s to now, artificial intelligence was significantly improved with the appearance of machine learning and deep learning in the study. It was able to perform not only decision-making but also predictions with outstanding outcomes in computer vision, recognition systems, and other domains. As a result, artificial intelligence becomes the research subject in order to apply it in many fields (Huang, 2024).

### 2.1.2 The Role of AI in Network Security Technologies

As shown in Figure 1 below, security in general is one of the fields where the application systems of artificial intelligence can be utilized to upgrade computer security technologies (Ma, 2023). Traditionally, these technologies in network security to protect data and prevent threats

are usually from setting up firewalls, configuring intrusion detection systems, and installing antivirus software. They work effectively toward their goals but require a lot of effort and cost in maintaining and updating the systems constantly. Moreover, they possibly become weaker and useless with new sorts of attacks. With applying artificial intelligence specifically machine learning and deep learning, these technologies become robust and automated. In other words, the network security systems can observe and study network traffic patterns, so they automatically accept or deny access without predefined settings. Furthermore, by studying attack patterns from the training data, the system like intrusion detection when enhanced with AI, can analyze and label unusual actions so that they identify and change the defense strategies against new types of attacks (Huang, 2024). Artificial intelligence plays an important role in network security as it makes technologies more intelligent and become agents that are able to perform the protection in real and constant time. Indeed, corporations and organizations can leverage it to secure their data and counter any network security threats.

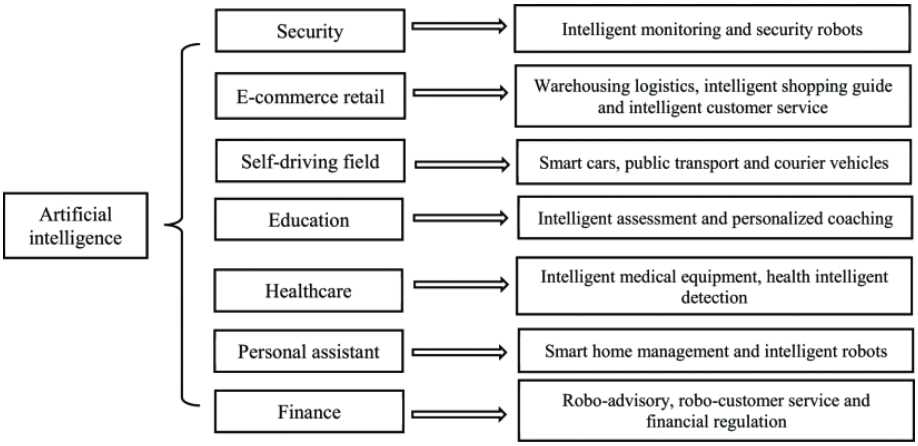


**Figure 1 – The potential fields can be applied by Artificial Intelligence. Adapted from "Application of artificial intelligence in computer network technology," by Ma, 2023**

## 2.2 External Threats

### 2.2.1 Background Information

External threats are a network security concern that corporations and organizations often encounter, and they can easily fall into traps due to computer security weaknesses. External threats refer to attacks that usually come from outside an enterprise's network. Threats are malware which are hard to recognize. According to Quinn, malware can be malicious software or computer programs that contain viruses, worms, and trojan horses. The author also mentioned ransomware which is a type of malware, and it is installed on computer systems by accidentally downloading the compromised software via online. Consequently, malware can exploit security holes and make the computer run slowly. Attackers will take advantage of accessing computer systems unauthorizedly, and they aim to steal, destroy, gain full control, and encrypt data and sensitive information to extort money (Quinn, 2020). As stated above, corporations usually use traditional computer security technologies, such as firewalls, intrusion detection systems, and malware detection like anti-virus software to fight against these external attacks. However, their effectiveness entirely depends on predefined rules and settings, and this requires lots of effort, time, and cost to maintain and update the information into the system.

### 2.2.2 The Application Models of AI in Network Security Technologies Solve External Threats

With the emergence of artificial intelligence, these technologies become robust and be capable of recognizing and blocking most attacks with high accuracy and fast computational process. For example, both Zhang et al (2021) and Han (2021) argued that the AI-based firewall is more intelligent and powerful compared to the traditional firewall. It can perform tasks exactly like the normal firewall. Moreover, it identifies and prevents malicious network traffic and malware by collecting and analyzing relevant data through probability and statistics. From these

methods, it helps to reduce the unnecessary workload of modifying predetermined settings while protecting data on computer systems and preventing incoming attacks effectively (Zhang, Chen, & Yang, 2021; Han, 2021). Besides, the application of artificial intelligence can be applied to intrusion detection system. Traditionally, the system can detect known threats or suspicious activity from network traffics based on predefined rules from the database. Applying a supervised machine learning algorithm such as Support Vector Machines (SVMs) enables the detection system to operate dynamically. Therefore, the system can perform the classification and prediction to a corporation's network traffic, and it will recognize novel activities automatically. Furthermore, by applying deep learning algorithms such as Convolutional Neural Networks (CNNs,) or Gradient Neural Networks (GNNs), the intrusion detection system is significantly enhanced. Detection systems use training datasets to analyze their characteristics, predict and identify attacks based on abnormal patterns and behaviors. As a result, the system is improved on precision, data processing to respond quickly, and the ability to detect both known and unknown malware. Like the firewall and the intrusion detection system, the malware detection traditionally detects blocks of code in known malicious software based on predefined libraries. With using AI models like deep learning algorithms, specifically CNNs and Recurrent Neural Network (RNNs), the ability of malware detection is upgraded in identifying and blocking malware. That means, the system can perform the analysis to not only the existing code but also abnormal behaviors or characteristics in malware (Huang, 2024).

## 2.3 Internal Threats

### 2.3.1 Background Information

Enterprises' data can become a target to internal threats. In contrast to external threats, internal threats usually originate from employees inside the corporation or organization. According to Nour et al, insiders who are associated with the enterprise play the role of threat

actors. There are two types: “authorized insiders,” who are employees privileged to access computer systems, and “opportunistic insiders,” who refer to former or current employees who do not have the privilege but use identity theft, pretending to someone’s identification, to gain the accessibility. The actions of bringing threats that result in damaging computer network security and impacting on data from insiders can be intentional and unintentional. With unintentional actions, employees may click on malicious buttons or spam emails while browsing websites, and this will directly bring malware to their corporation’s computer systems. On the other hand, with intentional actions, insiders involve in exploiting the weaknesses of computer security, executing the harmful malware to access organizations’ resources, and stealing or damaging them. It is hard to detect two types of patterns because employees may have knowledge about their enterprise’s security system, enabling them to make the security system temporarily weak or deactivated. As a result, employees may allow the external threats to enter computer systems undetected. Furthermore, employees take advantage of their authorization to simulate normal activities, so the security system cannot identify their suspicious behaviors to trigger an alert or prevention (Nour, Pourzandi, & Debbabi, 2023). Corporations and organizations usually have their own methods and policies to prevent internal threats so that they can protect their resources. For example, they will train their employees to acquire knowledge about network security and threats. Depending on each employee’s role in the corporation, configuring and accessing computer systems and resources are distinct to each individual. Besides, data and sensitive information are usually encrypted, protected with multi-factor authentication, and supervised by the highest authorized personnel.

### 2.3.2 The Application Models of AI in Network Security Solve Internal Threats

However, these techniques can be somehow evaded because of insiders as threat actors as described above; therefore, to maximize the effectiveness of these methods and minimize the impact of internal threats, organizations should focus on the prevention methods beforehand so that threats cannot happen. Nour et al in their survey paper provided one potential solution that combines artificial intelligence specifically machine learning algorithms with analytic methods. This solution is to mainly investigate data and information related to employees' activities from log files. It will detect and capture any abnormal behaviors or patterns to assist security technicians to identify and build up strategies against threats. There existed several designs based on this solution by utilizing supervised learning within the concept of machine learning.

For instance, the authors mentioned the framework designed by Alsaheel et al. to build an "end-to-end attack story" through audit log files by exploiting similarities across different attack patterns. This framework combines techniques of analysis, natural language processing, and machine learning algorithms to build a sequence-based model by using Long Short-Term Memory (LSTM) and CNNs. Its aim is to recognize patterns from attacks and to apprehend stealthy attack behaviors. Then, it will perform a prediction on attack probabilities by applying Dropout for regularization, Conv1D for processing the sequence, and a fully-connected layer with a sigmoid activation. Next, based on manual threat alerts, a causal graph is used to build a sequence diagram and point out abnormal nodes. The framework uses the sequence-based model to gather all nodes and identify ones that are linked to the contribution of the attack (Nour, Pourzandi, & Debbabi, 2023).

Villarreal-Vasquez et al.'s framework was to collect data from log events to detect internal threats from insiders. They build anomaly detection with LSTM by collecting event

sequences from multiple endpoints, and the operation of the method includes four parts such as data collection, data selection, model generation, and anomaly detection. After the phase of data collection, the method will use Endpoint Detection and Response (EDR) to choose relevant features from modifying the event feature factor. These features will contribute to performance results; therefore, if features are poorly selected, they impact the method's accuracy. From selected features, the framework will create the dataset used for training and validation. Based on the calculation of probabilities of different events, the method can learn normal event patterns and then identify anomalous activities (Nour, Pourzandi, & Debbabi, 2023).

The application systems of artificial intelligence in this context are like security robots since they can process a big amount of data related to activities in the short time, analyze attack patterns and abnormal behaviors, and help security specialists to identify and classify the threats. Indeed, it plays an important and useful role in preventing internal threats and protecting resources from insiders in corporations and organizations.

## 2.4 Challenges of AI in Network Security

Even though enterprises benefit from applying artificial intelligence into their network security technologies, they may also encounter several challenges. For example, most models of artificial intelligence need to collect and process a huge amount of data as training inputs for computers to learn. Depending on the field of the dataset, its content may contain personal information (e.g., full name, address, email, phone number, activities from browsing websites, etc.) and credential data (e.g., password, encrypted files, credit card information, etc.) Attackers will focus on these datasets and perform the threats to them. As a result, if corporations do not have methods for protection, they will face problems with "data privacy and security issues" related to not only datasets but also their resources (Tang, 2024). Without sample data, artificial

intelligence models can barely complete their jobs of automatically analyzing, recognizing, and preventing new attacks. Indeed, datasets are one of the important factors in the application systems of artificial intelligence, so the network security technologies can be more robust and intelligent as discussed above. However, by understanding the mechanism, attackers take this advantage to perform the tricks to mess up datasets by changing the contents or making them inaccurate. If the models use misleading data, learning is severely impacted, and results with low accuracy are produced. Thus, enterprises have to face the new threat defined as an "Adversarial Attack," as attackers exploit the weakness of Artificial Intelligence to counter AI-based network security technologies (Li, Zhang, Wang, & Xiao, 2023).

# 3. Methodology

## 3.1 Dataset

### 3.1.1 Overview

The dataset used for this project is named UNSW-NB15 dataset. It was contained by raw network traffic data and created and developed by Dr. Nour Moustafa through the application called IXIA PerfectStorm in UNSW Canberra, Australia. This dataset was split by two different files (i.e., the training set file and the testing set file). There are more than 175 thousand records in the training set, and there are approximately 83 thousand records in the testing set. Each record will be labeled as one out of nine types of attack, such as “Fuzzers”, “Analysis”, “Backdoors”, “DoS”, “Exploits”, “Generic”, “Recnnaissance”, “Shellcode”, and “Worms. Moreover, there are 49 features in each record including the class label. The training set file is preprocessed and used for training machine learning models, while the testing set file is used for checking the performance and comparing the accuracy among models. After that, the model with the highest accuracy score is selected to apply into the intrusion detection system. The preprocess steps are conducted by using the package named “pandas” and the Scikit-learn libraries like LabelEncoder, StandardScaler, OneHotEncoder, and ColumnTransformer.

### 3.1.2 Preprocessing and Normalizing Dataset

The dataset in both training and testing files is first preprocessed by dropping all invalid or empty records. Then, the process will continue dropping several features in records such as “id”, “srcip”, “sport”, “dstip”, and “dsport”. They are removed from the records because they contain sensitive or unnecessary data.

| **Features** | **Description** |
|---|---|
| id | The identification of each record |
| srcip | Source IP Address |
| sport | Source Port Number |
| dstip | Destination IP Address |

| | |
|---|---|
| dsport | Destination Port Number |

**Table 1 – The removed features and their description**

After doing the initial step, each record in the dataset just contains only 44 features left, and these features will be used for training the models.

```
        dur proto service state  spkts  dpkts  sbytes  dbytes       rate  \
0  0.121478   tcp       -   FIN      6      4     258     172  74.087490
1  0.649902   tcp       -   FIN     14     38     734   42014  78.473372
2  1.623129   tcp       -   FIN      8     16     364   13186  14.170161
3  1.681642   tcp     ftp   FIN     12     12     628     770  13.677108
4  0.449454   tcp       -   FIN     10      6     534     268  33.373826

   sttl  ...  ct_dst_sport_ltm  ct_dst_src_ltm  is_ftp_login  ct_ftp_cmd  \
0   252  ...                 1               1             0           0
1    62  ...                 1               2             0           0
2    62  ...                 1               3             0           0
3    62  ...                 1               3             1           1
4   254  ...                 1              40             0           0

   ct_flw_http_mthd  ct_src_ltm  ct_srv_dst  is_sm_ips_ports  attack_cat  \
0                 0           1           1                0      Normal
1                 0           1           6                0      Normal
2                 0           2           6                0      Normal
3                 0           2           1                0      Normal
4                 0           2          39                0      Normal

   label
0      0
1      0
2      0
3      0
4      0

[5 rows x 44 columns]
```

**Figure 2 – First few records of the dataset after an initial preprocessing step**

In order to be ready and available for training models, the dataset is needed to define the features and the target. Therefore, the inputs for training will have all features except for “label” and “attack_cat” as attack category. The purpose of the project is to analyze the network traffic data and classify them as either attack or normal, so the output as the target will hold the feature of “attack_cat”. Towards the output or target, its data type as a string is converted to the numerical type. Towards the inputs or features, they are split into two distinct data types: string and number for normalizing and scaling.

| Attack Category as String | Converted Numerical Type |
|---|---|
| Analysis | 0 |
| Backdoor | 1 |
| DoS | 2 |
| Exploits | 3 |
| Fuzzers | 4 |
| Generic | 5 |
| Normal | 6 |
| Reconnaissance | 7 |
| Shellcode | 8 |
| Worms | 9 |

**Table 2 – The String and Numerical Type of Each Attack Category**

## 3.2 Machine Learning Models

The primary purpose of this project is to use the AI-based intrusion detection system to monitor and classify the incoming network traffic as either normal or attack. Therefore, the machine learning models are mostly selected from the supervised ones, such as K-Nearest Neighbor (K-NN), Decision Tree, Support Vector Machine (SVM), Random Forest, and Extreme Gradient Boost (XGBoost). Each model is trained by using the preprocessed dataset and the Scikit-learn libraries. Here is the result of their performance after training.

| Model | Accuracy Score |
|---|---|
| K-Nearest Neighbor (K-NN) | 0.699 or 69.9% |
| Decision Tree | 0.698 or 69.8% |
| Support Vector Machine (SVM) | 0.684 or 68.4% |
| Random Forest | 0.754 or 75.4% |
| Extreme Gradient Boost (XGBoost) | 0.766 or 76.6% |

**Table 3 – The Performance of Each Model after Training**

According to the table, the model XGBoost has the highest accuracy score after training on the training records and classifying on the testing records. As a result, it is selected to apply into the intrusion detection system of this project.

## 3.3 Server Development

The backend side as the server was constructed by using the framework in Python like Flask. It included several Application Programming Interfaces (APIs) to receive and respond to the client's requests. The server was designed and implemented to handle the client's inputs from uploading files and manually entering network traffic flow data. By receiving the data from the user, the server would preprocess it by choosing appropriate features for model analysis. Then, it linked the client's processed data to the model Extreme Gradient Boost for prediction and classification. Finally, the server parsed the results from the model and returned them to the client as a response.

## 3.4 Client Development

The client was developed to mainly provide users with an environment to interact with the AI-based intrusion detection system through the user interface of the project. Therefore, they were built and implemented by using HTML and React library. Moreover, the design of their appearance would be completed by using Tailwind CSS. From the user interface in the client side, users can use input fields to upload files and manually enter the network data. There are some events, notifications, or warnings to handle or notify users about the inputs or errors. After sending the request for analyzing their network traffic input, users can visualize the result information from the model on the easily comprehensive dashboard page.

# 4. Results and Discussion

# 5. Conclusion

## 5.1 Future Work

## 5.2 Limitations

# 6. Reference


Mir, M. S., Wani, S., & Ibrahim, J. (2013). Critical information security challenges: An appraisal. *2013 5th International Conference on Information and Communication Technology for the Muslim World 2013 (ICT4M)*, 1–4. https://doi.org/10.1109/ICT4M.2013.6518890

J. Li, S. Zhang, J. Wang and H. Xiao, "Research on Artificial Intelligence Empowered Network Security," 2023 9th International Conference on Computer and Communications (ICCC), Chengdu, China, 2023, pp. 1267-1271, doi: 10.1109/ICCC59590.2023.10507302.

Huang, H. (2024). Analysis of the Application of Artificial Intelligence in Computer Network Technology. *Studies in Social Science Research*, *5*(3), p13. https://doi.org/10.22158/sssr.v5n3p13

X. Ma, "Application of artificial intelligence in computer network technology," 2023 2nd International Conference on Artificial Intelligence and Autonomous Robot Systems (AIARS), Bristol, United Kingdom, 2023, pp. 182-186, doi: 10.1109/AIARS59518.2023.00043.

Quinn, M. J. (2020). Ethics for the information age (8th ed.). Pearson.

Zhang, lei, Chen, Z., & Yang, S. (2021). Application of artificial intelligence in computer network security. *Journal of Physics. Conference Series*, *1865*(4), 42039. https://doi.org/10.1088/1742-6596/1865/4/042039

Z. Han, "The Application of Artificial Intelligence in Computer Network Technology," 2021 2nd International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT), Shanghai, China, 2021, pp. 632-635, doi: 10.1109/AINIT54228.2021.00127.

B. Nour, M. Pourzandi and M. Debbabi, "A Survey on Threat Hunting in Enterprise Networks," in IEEE Communications Surveys & Tutorials, vol. 25, no. 4, pp. 2299-2324, Fourthquarter 2023, doi: 10.1109/COMST.2023.3299519

Q. Tang, "Exploration of the Application of Artificial Intelligence in Computer Networks," 2024 3rd International Conference on Artificial Intelligence and Autonomous Robot Systems (AIARS), Bristol, United Kingdom, 2024, pp. 50-55, doi: 10.1109/AIARS63200.2024.00015

N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in Proc. Mil. Commun. and Inf. Syst. Conf. (MilCIS), 2015, pp. 1-6.

N. Moustafa and J. Slay, "The evaluation of Network Anomaly Detection Systems: Statistical analysis of the UNSW-NB15 dataset and the comparison with the KDD99 dataset," Inf. Secur. J.: A Global Perspect., pp. 1-14, 2016.

N. Moustafa et al., "Novel geometric area analysis technique for anomaly detection using trapezoidal area estimation on large-scale networks," IEEE Trans. Big Data, 2017.

N. Moustafa et al., "Big data analytics for intrusion detection system: statistical decision-making using finite dirichlet mixture models," in Data Analytics and Decision Support for Cybersecurity, Cham, Switzerland: Springer, 2017, pp. 127-156.

M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow Datasets for Machine Learning-Based Network Intrusion Detection Systems," in Proc. 10th EAI Int. Conf. Big Data Technol. Appl. (BDTA), Dec. 2020, p. 117.